

Solution Requirements (Functional & Non-functional)

Date	28 June 2026
Team ID	XXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	4Marks

Functional Requirements (FR):

Following are the functional requirements of the proposed solution.

FR No.	Category	Requirement Description	Priority
1	User Interface	The system must provide an intuitive web interface that allows users to select a topic, upload an audio file, or record their voice directly for concept evaluation.	High
2	Speech Recognition	The system must accurately convert the user's spoken explanation into text using the Whisper Speech Recognition model before further analysis.	High
3	NLP Processing	The system must compare the generated transcript with the reference concept using Sentence-BERT semantic similarity to evaluate conceptual understanding rather than keyword matching.	High
4	AI Evaluation	The system must calculate similarity score, grammar score, understanding level, confidence rating, and overall performance automatically after processing the audio.	High
5	Reporting	The system must generate a downloadable PDF evaluation report containing the user's performance metrics, AI feedback, matched words, missing concepts, and improvement suggestions.	Medium

6	Certificate Generation	The system must automatically generate a personalized certificate for users who successfully complete the concept evaluation with satisfactory performance.	Medium
7	Business Rules	The system must evaluate only supported audio formats, maintain reference text consistency for each selected topic, and ensure evaluation results are generated only after successful speech transcription.	High

Non-Functional Requirements (NFR):

Following are the non-functional requirements of the proposed solution.

NFR No.	Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application should process uploaded audio efficiently and display AI evaluation results with minimal waiting time.	Speech transcription and similarity evaluation should complete within 10–20 seconds for normal audio recordings.
2	Scalability	The application should support evaluation of multiple concepts and allow future integration of additional subjects and users.	The modular architecture should allow adding new topics, reference texts, and AI models without major code modifications.
3	Security & Data Privacy	User audio recordings and evaluation reports should be handled securely without exposing personal information.	Uploaded audio should only be used for evaluation, and reports should not disclose sensitive user information.
4	Reliability & Availability	The application should consistently generate accurate evaluation results and remain stable during continuous usage.	The system should successfully process supported audio files and produce reliable evaluation reports without unexpected failures.

5	Usability& Accessibility	The application should provide a simple, responsive, and easy-to-understand interface suitable for students and faculty members.	Users should be able to complete the entire evaluation process without technical assistance, with clearly labeled controls and understandable feedback.
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